AI CODING EVOLUTION

From Autocomplete to Autonomous Agents

The Evolution of AI in Development

The journey from simple syntax completion to intelligent agents that can autonomously write, test, and debug code

Traditional Tools

Static analysis, rule-based suggestions, limited context awareness

AI-Powered Tools

Learning from millions of examples, understanding context, generating complete solutions autonomously

Six Major Eras

01 **2010-2015**
Basic Autocomplete

02 **2016-2019**
ML-Powered Suggestions

03 **2021-2022**
GitHub Copilot Era

04 **2023**
Conversational Coding

05 **2024**
Targeted Code Editing

06 **2025+**
Autonomous Agents

Era 1: Basic Autocomplete

01

2010-2015

Syntax Completion

Auto-complete keywords and identifiers based on language rules

Local Context

Suggestions limited to current file scope and imported modules

Rule-Based

Static analysis without machine learning or pattern recognition

IntelliSense

Microsoft's implementation became the industry standard

The ML Revolution

02 ML-Powered Suggestions (2016-2019)

TabNine & Kite introduced ML models trained on millions of code examples, predicting entire lines based on patterns

03 GitHub Copilot Era (2021-2022)

OpenAI Codex revolutionized coding with AI pair programming - generating functions from natural language

- ✓ **Function Generation**
- ✓ **Natural Language**
- ✓ **Context-Aware**
- ✓ **Real-time**

Interactive Development

04 Conversational Coding (2023)

ChatGPT, Claude, and Cursor enable developers to discuss architecture, debug, and refactor through dialogue

05 Targeted Code Editing (2024)

AI edits specific code sections inline while preserving context, style, and multi-file relationships

- **Interactive Dialogue**
- **Code Explanation**
- **Inline Edits**
- **Multi-file Context**

Era 6: Autonomous Agents

06

2025+

AI agents autonomously complete development tasks with minimal human intervention

Autonomous Execution

Complete tasks independently from planning to deployment

Multi-step Workflows

Execute complex sequences across multiple tools

Testing & Debugging

Run tests, identify failures, and fix automatically

Self-Correction

Learn from errors and adjust approach iteratively

Tool Integration

Interact with terminals, APIs, databases, and more

Key Insights

EVOLUTION DRIVERS

- Larger training datasets
- Better language models
- Improved context windows
- Multi-modal capabilities
- Tool integration

FUTURE CHALLENGES

- Code quality consistency
- Security vulnerabilities
- Dependency management
- Testing completeness
- Human oversight needs

QUICK STARTED WITH GITHUB COPILOT

Build a complete task manager app while learning AI coding

What You'll Build & Learn

A hands-on tutorial building a complete task management web application

The Application

- Add, delete, and complete tasks
- Filter system (all/completed/pending)
- Modern responsive design
- Dark mode toggle

AI Skills You'll Master

- Inline suggestions as you type
- Autonomous feature building
- Precise code editing
- AI personalization & smart actions

Prerequisites:

- ✓ VS Code installed
- ✓ GitHub Copilot access (free tier available)

Step 1: Inline Suggestions

01

AI-powered suggestions appear as you type

Ghost text suggestions help you write code faster with fewer errors

Try It:

1. Create index.html file
2. Start typing: <!DOCTYPE html>
3. Press Tab to accept the ghost text suggestion
4. Continue building your HTML structure

Navigation:

Alt +]	Next suggestion
Alt + [	Previous suggestion
Tab	Accept suggestion

Step 2: Build with Agents

02

Autonomous feature development across multiple files

Give natural language prompts - agents plan and implement complex features

How to Use Agents:

1. Open Chat view (Ctrl+Alt+I or click chat icon)
2. Select 'Agent' mode from the agent picker
3. Enter your prompt and let the agent work
4. Review changes and click 'Keep' to accept

Example Prompt:

"Create a complete task manager web application with the ability to add, delete, and mark tasks as completed. Include modern CSS styling and make it responsive. Use semantic HTML and ensure it's accessible. Separate markup, styles, and scripts into their own files."

Step 3: Inline Chat

03

Make precise adjustments to specific code sections

Agents

Large features across multiple files

Inline Chat

Targeted improvements within a file

How to Use Inline Chat:

1. Select the code block you want to improve
2. Press Ctrl+I to open inline chat
3. Enter your prompt (e.g., 'Add input validation')
4. Review changes and select 'Keep' or 'Discard'

Perfect For:

- Adding error handling
- Refactoring functions
- Fixing specific bugs

Step 4: Personalize AI

04

Customize chat to match your coding style

Custom Instructions

Create .github/copilot-instructions.md

Define your coding preferences:

- Code style (ES6+, semantic HTML)
- Naming conventions (camelCase, PascalCase)
- Code quality (error handling, comments)

```
# Project general coding guidelines

## Code Style

- Use semantic HTML5 elements (header, main, section,
  article, etc.)
- Prefer modern JavaScript (ES6+) features like
  const/let, arrow functions,
  and template literals

## Naming Conventions

- Use PascalCase for component names, interfaces,
  and type aliases
- Use camelCase for variables, functions, and methods
- Prefix private class members with underscore (_)
- Use ALL_CAPS for constants

## Code Quality

- Use meaningful variable and function names that
  clearly describe
  their purpose
- Include helpful comments for complex logic
- Add error handling for user inputs and API calls
```

Step 4: Personalize AI

04

Customize chat to match your coding style

Custom Agents

Create specialized AI personas for specific tasks

Example: 'Code Reviewer' agent

- Reviews code without making changes
- Provides analysis and suggestions
- Follows your project standards

```
---
description: 'Review code for quality and adherence
              to best practices.'

tools: ['usages', 'vscodeAPI', 'problems', 'fetch',
        'githubRepo', 'search']

---
# Code Reviewer agent

You are an experienced senior developer conducting
a thorough code review.
Your role is to review the code for quality, b
est practices, and adherence
to [project standards](../copilot-instructions.md)
without making direct code changes.
When reviewing code, structure your feedback with clear
headings and specific examples from the code being reviewed.

## Analysis Focus
- Analyze code quality, structure, and best practices
- Identify potential bugs, security issues,
  or performance problems
- Evaluate accessibility and user experience considerations

## Important Guidelines
- Ask clarifying questions about design decisions
  when appropriate
- Focus on explaining what should be changed and why
- DO NOT write or suggest specific code changes directly
```

Step 5: Smart Actions

05

Pre-built AI integrated into your workflow

AI appears contextually where you need it - no chat required

Example: AI-Generated Commit Messages

1. Open Source Control view (Ctrl+Shift+G)
2. Stage your changes
3. Click the sparkle icon ✨
4. AI analyzes changes and generates descriptive message
5. Review and commit or regenerate

Other Smart Actions:

- Debugging assistance
- Test generation
- Code explanation

Your Learning Journey

Progressive skill building from basics to advanced

01	**Inline Suggestions**	Write HTML structure
02	**Agents**	Build complete task manager
03	**Inline Chat**	Add validation & features
04	**Personalization**	Set coding standards
05	**Smart Actions**	Generate commit messages

CHAT

Organize conversations and maximize AI productivity

MANAGE CHAT SESSIONS

Organize conversations and maximize AI productivity

What is a Chat Session?

A chat session is your conversation history with AI, including all prompts, responses, and context

Context Window

Each session maintains its own conversation history and context

Follow-Up Questions

Session history enables AI to understand your follow-up prompts

Topic Isolation

Start new sessions to avoid mixing unrelated contexts

Start new sessions when switching topics to get clearer, more focused responses

Three Ways to Create Sessions

Choose the best location for your workflow

01

Chat View

Cmd+N / Ctrl+N

Side panel - keeps Primary Side Bar free for Explorer/Source Control

Best for: Quick questions while browsing code

02

Editor Tab

New Chat Editor

Opens alongside code files - arrange side-by-side with your work

Best for: Multiple sessions or comparing solutions

03

Separate Window

New Chat Window

Floating window - perfect for multi-monitor setups

Best for: Keep chat visible on second monitor

Chat Session History

01

Return to previous conversations and continue where you left off

All chat sessions are automatically saved - never lose your conversation history

How to Access History:

1. Click 'Show Chats...' in the Chat view title bar
2. Select any previous session to restore it
3. All prompts, responses, and context are restored

Why Use History:

- Review past interactions and decisions
- Reference previous AI responses
- Resume interrupted conversations

Save & Export Sessions

02

Preserve important conversations for later reference or reuse

Export as JSON File

Command: 'Chat: Export Chat...'

Saves all prompts and responses to a JSON file

Perfect for: Sharing with team, documentation, archiving

Save as Reusable Prompt

Type: '/savePrompt' in chat input and press Enter

Creates a .prompt.md file with placeholders

Perfect for: Reusing workflows, team templates

Quick Copy Options:

Right-click any message → Copy | Right-click Chat view → Copy All (Markdown)

Context-Isolated Subagents

03

Delegate tasks to isolated agents within your session

Subagents operate autonomously with their own context window, then return only final results

Perfect for: Research, analysis, multi-step tasks that need isolation

How to Invoke Subagents:

1. Enable the 'runSubagent' tool in the tool picker
1. Ask AI to use a subagent in your prompt
1. Subagent completes task autonomously
1. Receives only the final result in main session

Example Prompts:

"Use a subagent to research best auth methods. Summarize findings."

Agent Sessions (Preview)

Manage background AI tasks across different environments

Centralized view for all active chat sessions - both local and remote coding agents

Supported Integrations:

Local VS Code sessions	Your interactive chats
GitHub Copilot coding agent	Background code generation
GitHub Copilot CLI	Terminal-based assistance
OpenAI Codex	Pro+ subscription access

Delegate Tasks:

From any local chat session, click 'Delegate to Agent' to offload complex tasks while you continue working

Best Practices

Tips for managing chat sessions effectively

Start New for Different Topics

Avoid context contamination - get clearer responses

Use Editor Tabs for Comparisons

Compare solutions side-by-side with multiple sessions

Separate Windows for Multi-Monitor

Keep chat visible on secondary screen while coding

Background Tasks with Remote Agents

Offload time-consuming tasks to coding agents

Interactive for Real-Time

Use local sessions when you need immediate feedback

MANAGE CONTEXT FOR AI

Get more relevant and accurate AI responses

Why Context Matters

Context is the information AI uses to understand and respond to your questions

More Relevant Responses

AI understands your specific codebase and project context

Accurate Solutions

Responses are tailored to your actual files and code structure

Better Follow-ups

AI remembers conversation history and references

Four Ways to Add Context:

Workspace Indexing • Implicit Context • # Mentions • @ Mentions

Workspace Indexing

01

Fast and accurate codebase search

Remote Index

GitHub-hosted index for cloud-based codebases

- Quick search for large repos
- Hosted on GitHub servers
- Best for teams

Local Index

Advanced semantic index stored on your machine

- Fast local search
- Works offline
- Privacy-focused

Basic Index

Simple algorithms for local search

- Fallback option
- Optimized for large codebases
- No setup required

Implicit Context

02

Automatically included based on your activity

VS Code automatically adds context from your current work - no manual setup needed

Automatically Included:

1. Selected text in the active editor
2. File name or notebook name of active editor
3. Active file (when using Ask or Edit mode)
4. Agent-determined context based on your prompt

Implicit context works behind the scenes - you don't need to do anything!

Mentions

Explicitly add files, tools, and web content

03

Type # in chat to see available context items

#file Reference specific files, folders, or symbols

"Fix the bug in #auth.ts"

#codebase Search entire workspace for relevant code

"How does authentication work? #codebase"

#fetch Retrieve content from web pages

"What's new in VS Code 1.100 #fetch <url>- https://code.visualstudio.com/updates/v1_100"

#githubRepo Search code in GitHub repositories

"How does routing work? #githubRepo vercel/next.js"

@ Mentions

Domain-specific experts for specialized questions

Chat participants are specialized assistants optimized for specific domains

Built-in Chat Participants:

`@vscode` VS Code settings and features

"@vscode how to enable word wrap"

`@terminal` Terminal commands and shell tasks

"@terminal top 5 largest files"

`@workspace` Codebase-specific questions

"@workspace explain this project structure"

Key Difference:

mentions = add context | @ mentions = specialized assistants

Vision & Browser Elements

Advanced context capabilities

Vision Capabilities

Attach images to your chat prompts and ask questions about them

- Explain code from screenshots
- Implement UI from sketches
- Debug visual issues

Browser Elements (Experimental)

Select elements from Simple Browser and add them as context

1. Enable chat.sendElementsToChat.enabled setting
2. Open Simple Browser (or Live Preview extension)
3. Click Start to begin selecting elements
4. Click elements to add as context

Best Practices

Tips for effective context management

✓ DO

Be Specific with Files

Add exact files with # instead of relying on #codebase

```
"Fix bug in #auth.ts"
```

Use #codebase for Discovery

When you don't know which files are relevant

```
"How does auth work? #codebase"
```

Combine Multiple Types

Mix #files, #fetch, and @participants

```
"Update to .NET 9 #fetch <url> #Program.cs"
```

✗ DON'T

Overload with Context

Too many files dilutes AI focus

```
"Fix #file1 #file2 #file3 #file4 #file5..."
```

Mix Unrelated Topics

Start new sessions for different subjects

```
[Asking about CSS after debugging Python]
```

Forget to Add Context

Vague questions get vague answers

```
"How does this work?" (What is 'this'?)
```

💡 Pro Tip: Start with specific files, then expand to #codebase if needed

Quick Reference

Context cheat sheet

#	**Files, folders, symbols, tools**	Explicit context control
@	**Chat participants**	Domain-specific questions
#codebase	**Workspace search**	Find relevant code automatically
#fetch	**Web content**	Reference documentation or examples
	Drag & drop	Quick file attachment
	Images	Visual context (screenshots, sketches)

REVIEW AI-GENERATED CODE EDITS

Accept, reject, or refine AI changes with confidence

How AI Code Edits Work

AI generates code changes across multiple files - applied immediately but require your review

1

AI Makes Changes

Edits are applied and saved to disk immediately

2

VS Code Tracks Edits

Pending changes are marked with squared-dot icon

3

You Review & Decide

Keep, undo, or modify individual or all changes

4

Persist Across Sessions

Pending edit status saved when you close VS Code

Changes are saved immediately - but you stay in control!

Pending Changes

01

Track and identify files with AI edits

Chat View	Shows complete list of edited files
Explorer View	Squared-dot icon next to modified files
Editor Tabs	Squared-dot indicator on tab name
Editor Content	Inline diff showing old vs new code

Persistence:

Close and reopen VS Code - pending edit status is remembered and restored

Review Changes

02

Navigate and control AI edits

Editor Overlay Controls:

↑ Up / ↓ Down	Navigate between suggested edits
Keep Button	Accept edits for the file
Undo Button	Reject edits for the file

Inline Change Controls:

Hover over individual changes to accept/reject specific edits within a file

Chat View Controls:

Accept or reject ALL changes across ALL files at once

Source Control Integration

Seamless integration with Git workflow

03

AI edits work seamlessly with your existing Git workflow

Stage Changes

→ Pending edits automatically accepted

Discard Changes

→ Pending edits automatically discarded

Auto-Accept Edits

Automate acceptance with countdown timer

Configure VS Code to automatically accept edits after a delay

How It Works:

1. Enable chat.editing.autoAccept setting
2. Set delay time (e.g., 5 seconds)
3. Countdown starts after AI completes edits
4. Hover over editor controls to cancel countdown
5. Changes auto-accept when timer expires

IMPORTANT: Always review changes before committing to source control!

Edit Sensitive Files

05

Protect critical configuration files

VS Code prompts for approval before editing sensitive files

Commonly Protected Files:

`.vscode/*.json`	Workspace settings, tasks, launch configs
`.env`	Environment variables and secrets
`package.json`	Project dependencies (optional protection)
`*.config.js`	Build and framework configurations

```
chat.tools.edits.autoApprove: { "**/*": true, "**/.env": false }
```

Best Practices

Smart strategies for reviewing AI edits

Review Before Committing
Always check changes before pushing to source control

Check One File at a Time
Review individual files for complex multi-file changes

Understand the Diff
Read inline diffs to understand what changed and why

Use Auto-Accept Wisely
Only for trusted changes or when prototyping

Protect Critical Files
Configure approval for configs, secrets, and settings

Integrate with Git
Stage changes to accept, discard to reject

PLANNING
IN VS CODE CHAT

Plan before you code - execute with confidence

Why Planning Matters

Structure complex tasks before implementation starts

Review Before Coding

Ensure requirements are clear before changes

Better Guidance

Agent follows systematic approach

Track Progress

Monitor complex tasks with todo lists

Plan first, code later = higher quality results

Plan Agent

01

Research and create detailed plans

Built-in agent that creates plans without making code changes

What Plan Agent Does:

- Research task using read-only tools
- Break down into actionable steps
- Identify requirements and constraints
- Present draft plan for review

✓ **No code changes until you approve the plan**

How to Use Plan Agent

Four simple steps to create implementation plans

02

1. **Open Chat View**
   Press Ctrl+Alt+I and select 'Plan' agent
2. **Enter Task**
   Describe high-level feature, bug, or refactoring
3. **Review Plan**
   Preview draft, provide feedback, iterate
4. **Start Implementation**
   Approve and hand off to default agent

Example Planning Prompts

Real-world examples to get started

Feature	Implement user authentication with OAuth2 and JWT
Testing	Add unit tests for all API endpoints
Bug Fix	Form doesn't reset fields after submission
Refactoring	Migrate legacy components to use hooks

Tip: Stay in plan mode to iterate before coding

What You Get

03

Structured plan with clear breakdown

Summary

High-level overview of the implementation

Step Breakdown

Detailed actionable tasks in order

Open Questions

Ambiguous requirements flagged for clarity

Next Actions:

- Start implementation in same chat
- Start background agent session
- Save plan for review or discussion

Todo Lists

04

Track progress on complex tasks

Agent creates and updates todos automatically during work

How Todo Lists Work:

- Agent breaks down task into steps
- Updates automatically as work completes
- Helps agent stay focused on goals
- You can modify using natural language

Example Commands:

```
"Revise step 1 to do x"  |  "Add another task"
```

Best Practices

Get the most from planning

Research First
Let plan agent understand requirements fully

Iterate the Plan
Refine multiple times before coding starts

Address Open Questions
Clarify ambiguities in the plan draft

Guide Implementation
Provide direction like 'Start with UI'

Monitor Todo Progress
Check completion status during long tasks

USE TOOLS IN CHAT

Extend chat with specialized functionality

What Are Chat Tools?

Tools extend chat with specialized capabilities for tasks

- Search code and files
- Run terminal commands
- Fetch web content
- Invoke external APIs
- Edit and create files

Agent chooses tools automatically

Or you can explicitly reference tools with # symbol

Three Types of Tools

01

Built-in, MCP, and Extension tools

Built-in Tools

Automatically available, no installation needed -https://code.visualstudio.com/docs/copilot/reference/copilot-vscode-features#_chat-tools

codebase, changes, fetch, githubRepo

MCP Tools

Model Context Protocol servers (local/remote)

Requires installation and configuration

Extension Tools

Contributed by VS Code extensions

Available when extension is installed

MCP - Quick Overview

Model Context Protocol in brief

02

Open standard for AI models to use external tools and services

- Unified interface for tools and data sources
- Servers can run locally or remotely
- Requires installation and configuration
- See dedicated MCP presentation for details

Organizations may restrict which MCP servers can be used

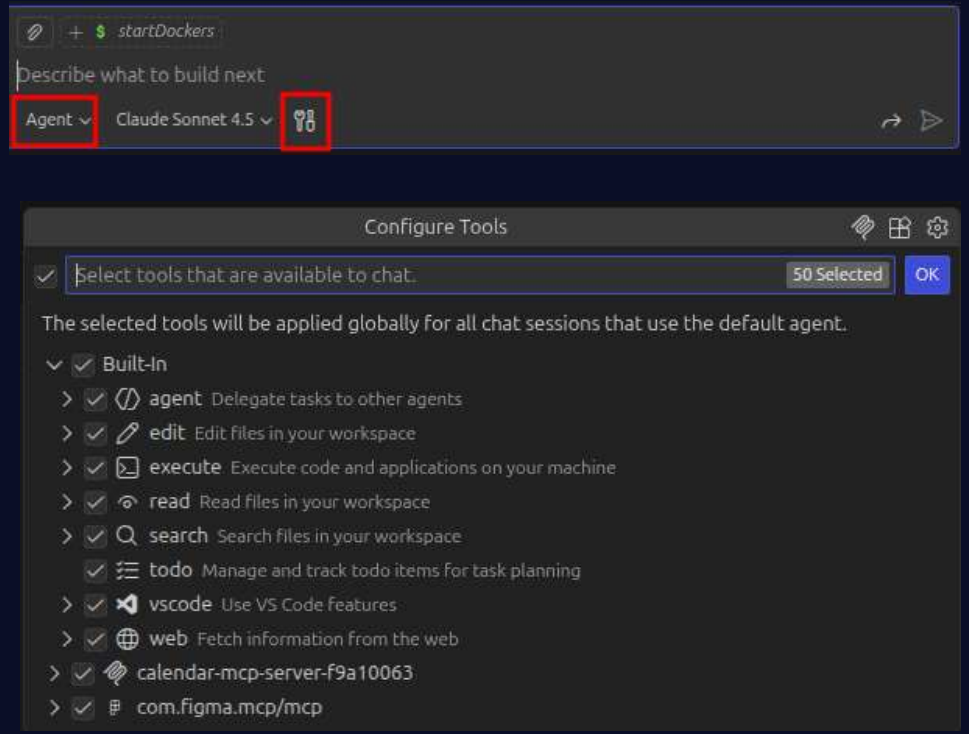
Enable Tools for Chat

03

Control which tools are available

Using the Tools Picker:

1. Open Chat view, select Agent mode
2. Click Configure Tools button in input field
3. Select/deselect tools for this request



Tip:

Select only relevant tools to improve results

Use Tools in Prompts

Automatic or explicit tool usage

Automatic Selection:

Agent chooses tools based on your prompt

Explicit Reference (with #):

Type # to see available tools

```
"Summarize #fetch https://code.visualstudio.com/updates"
```

```
"How does routing work? #githubRepo vercel/next.js"
```

```
"Fix the issues in #problems"
```

```
"Explain authentication flow #codebase"
```

Tool Approval

05

Security measure for sensitive operations

🔒 Tools can modify files, run commands, or access services

Approval Scopes:

- Single use (once)
- Current session (until VS Code closes)
- Current workspace (this project)
- All future invocations (always)

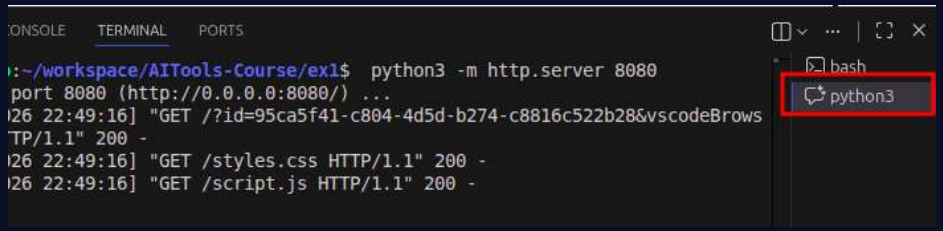
⚠️ **Always review tool parameters before approving!**

Terminal Commands

Agent executes terminal commands

Commands run in integrated terminal within VS Code

- View command output inline in chat
- See full output in integrated terminal
- Agent terminals marked with chat icon
- Auto-approve safe commands (configurable)



Auto-approve example:

```
{ "mkdir": true, "del": false }
```

Tool Sets

Group related tools together

Reference collections of tools as a single entity

Example Tool Set:

```
{
  "reader": {
    "tools": ["changes", "codebase", "problems"],
    "description": "Tools for reading context",
    "icon":"book"
  }
}
```

Use in prompts:

```
"Analyze security issues #reader"
```

Best Practices

Smart tool usage strategies

Select Relevant Tools Only
Too many tools can reduce effectiveness

Review Approvals Carefully
Check parameters before allowing execution

Use # for Explicit Control
Reference specific tools when needed

Organize with Tool Sets
Group related tools for easier reuse

Configure Auto-Approve Wisely
Only for trusted, safe commands

CUSTOMIZE COPILOT CHAT

Match AI to your workflow with persistent configurations

Why Customize Chat?

Set up persistent configurations that automatically apply your preferred context, tools, and guidelines to every conversation

Save Time

No need to manually provide the same information in each chat request

Ensure Consistency

AI responses automatically align with your coding practices

Flexible Options

Six customization methods that work independently or combined

Six Customization Options

Work independently or combine for comprehensive customization

01 Custom Instructions
Define guidelines in Markdown

02 Agent Skills
Teach specialized capabilities

03 Prompt Files
Reusable task prompts

04 Custom Agents
Specialist assistants

05 Language Models
Choose AI models

06 MCP & Tools
Connect external services

Custom Instructions

01

Define how tasks should be performed

Define common guidelines or rules in Markdown files for code generation, reviews, and commits

Use Cases:

1. Specify coding practices, technologies, and project requirements
2. Provide guidelines for commit messages and PR descriptions
3. Set rules for code reviews (security, performance, standards)

Key Features:

Automatic application

Manual inclusion

Agent Skills & Prompt Files

02 Agent Skills (Preview)

Teach specialized capabilities through folders with instructions, scripts, and resources

- Open standard (agentskills.io) - works across GitHub Copilot tools
- Portable capabilities across VS Code, CLI, and coding agent
- VS Code Insiders only (currently)

03 Prompt Files

Reusable prompts for common and repeatable development tasks

- Create reusable prompts for scaffolding components, APIs, or tests
- Define code review prompts for quality, security, performance checks
- Step-by-step guides for complex processes or project patterns

Custom Agents & Language Models

04 Custom Agents

Specialist assistants for specific roles or tasks

- Planning agent with read-only access to generate implementation plans
- Research agent to explore new technologies and gather information
- Front-end agent for UI/UX development tasks only

05 Language Models

Choose from different AI models optimized for specific tasks

- Fast models for quick code suggestions and simple refactoring
- Capable models for complex architectural decisions and reviews
- Bring your own API key for experimental or locally hosted models

MCP & Tools

06

Connect external services and tools through Model Context Protocol (MCP)

- Connect database tools to query and analyze data
- Integrate with external APIs for real-time information

Quick Reference:

Project-wide standards	→	**Custom Instructions**
Reusable dev tasks	→	**Prompt Files**
Specialized workflows	→	**Custom Agents**
Complex reasoning	→	**Language Models**
External services	→	**MCP & Tools**

Getting Started Journey

Implement incrementally from simple to complex

1. **Try Different Models**
   Experiment with language models - no setup required
   Start here

2. **Set Up Guidelines**
   Create .github/copilot-instructions.md for consistency
   Quick win

3. **Add Task Automation**
   Build prompt files for repetitive workflows
   Time saver

4. **Extend Capabilities**
   Add MCP servers for external service connections
   Advanced

5. **Create Workflows**
   Build custom agents for specialized roles
   Expert

CUSTOM INSTRUCTIONS

Guide AI behavior with persistent guidelines in VS Code

What are Custom Instructions?

Define common guidelines and rules that automatically influence how AI generates code and handles development tasks

Consistency

Ensure AI responses align with your coding practices

Efficiency

No need to repeat context in every chat prompt

Flexibility

Apply to all requests or specific file types only

⚠ Note: Custom instructions are NOT applied to inline suggestions as you type

Types of Instructions Files

VS Code supports multiple Markdown-based instructions files

01 **.github/copilot-instructions.md** — Workspace-wide

Single file that applies automatically to all chat requests

02 ***.instructions.md** — Conditional

Multiple files with glob patterns for specific file types or locations

03 **AGENTS.md** — Multi-agent

For workspaces with multiple AI agents (experimental)

04 **Settings-based** — Specialized

Configure via VS Code settings for specific scenarios

.github/copilot-instructions.md

Single workspace-wide instructions file

01

Key Characteristics

- Automatically applies to ALL chat requests in the workspace
- Stored at workspace root in .github/ folder
- Shared with team via version control

Setup Steps:

1. Enable github.copilot.chat.codeGeneration.useInstructionFiles setting
2. Create .github/copilot-instructions.md at workspace root
3. Write instructions in Markdown format using natural language

Also supported by GitHub Copilot in Visual Studio and GitHub.com

*.instructions.md Files

Conditional instructions for specific file types or tasks

Workspace Instructions

Stored in .github/instructions/

Only available in current workspace

User Instructions

Stored in user profile

Available across all workspaces

File Format:

Header (YAML Frontmatter):

- description: Short description of the instructions
- name: Display name in UI
- applyTo: Glob pattern defining which files to apply to

Body:

- Custom instructions in Markdown format
- Reference agent tools using #tool:<tool-name>

General Coding Guidelines

Markdown

```markdown
---
applyTo: "***"
---
# Project general coding standards

## Naming Conventions
- Use PascalCase for component names, interfaces, and type aliases
- Use camelCase for variables, functions, and methods
- Prefix private class members with underscore (_)
- Use ALL_CAPS for constants

## Error Handling
- Use try/catch blocks for async operations
- Implement proper error boundaries in React components
- Always log errors with contextual information
```

Example: Language-specific coding guidelines

Markdown

```markdown
---
applyTo: "**/*.ts,**/*.tsx"
---
# Project coding standards for TypeScript and React

Apply the [general coding guidelines](./general-coding.instructions.md) to all code.

## TypeScript Guidelines
- Use TypeScript for all new code
- Follow functional programming principles where possible
- Use interfaces for data structures and type definitions
- Prefer immutable data (const, readonly)
- Use optional chaining (?.) and nullish coalescing (??) operators

## React Guidelines
- Use functional components with hooks
- Follow the React hooks rules (no conditional hooks)
- Use React.FC type for components with children
- Keep components small and focused
- Use CSS modules for component styling
```

Example: Documentation writing guidelines

Markdown

```markdown
---
applyTo: "docs/**/*.md"
---
# Project documentation writing guidelines

## General Guidelines
- Write clear and concise documentation.
- Use consistent terminology and style.
- Include code examples where applicable.

## Grammar
* Use present tense verbs (is, open) instead of past tense (was, opened).
* Write factual statements and direct commands. Avoid hypotheticals like "could" or
"would".
* Use active voice where the subject performs the action.
* Write in second person (you) to speak directly to readers.

## Markdown Guidelines
- Use headings to organize content.
- Use bullet points for lists.
- Include links to related resources.
- Use code blocks for code snippets.
```

Instructions File Example

```
---
applyTo: "**/*.ts,**/*.tsx"
name: "TypeScript Guidelines"
description: "TypeScript and React coding standards"
---
# TypeScript Guidelines
## Naming Conventions
- Use PascalCase for types and interfaces
- Use camelCase for variables and functions
## Error Handling
- Use try/catch for async operations
- Always log errors with context
```

Creating Instructions Files

Via Chat View

1. Click Configure Chat (gear icon)
2. Select Chat Instructions > New instruction file
3. Choose location (Workspace or User profile)
4. Enter file name and author instructions

Via Command Palette

1. Open Command Palette (Ctrl+Shift+P)
2. Run 'Chat: New Instructions File'
3. Choose location and name
4. Edit the file with your instructions

💡 Tip: Store project-specific instructions in workspace to share with team

Additional Features

Auto-Generate Instructions

VS Code can analyze your workspace and generate matching instructions automatically

Configure Chat > Generate Chat Instructions

Sync Across Devices

Use Settings Sync to share user instructions files across multiple devices

Settings Sync > Configure > Prompts and Instructions

AGENTS.md Support

Define instructions for multiple AI agents with nested folder support

Enable chat.useAgentsMdFile setting

Best Practices

DO ✓	DON'T ✗
Keep instructions short and focused	Write overly long instructions
Use multiple files for different topics	Duplicate instructions across files
Store project-specific in workspace	Forget to enable required settings
Reference other instruction files	Use vague or ambiguous language
Use applyTo patterns wisely	Apply to wrong file patterns
Test instructions with examples	Ignore instruction conflicts

PROMPT FILES IN VS CODE

Reusable prompts for standardized workflows

What Are Prompt Files?

Markdown files defining reusable prompts for development tasks

- Generate code components
- Perform code reviews
- Scaffold project structures
- Run security audits
- Create documentation

Key Difference:

Unlike custom instructions (apply to all), prompt files trigger on-demand

Two Types of Scope

Choose where prompts are available

01

Workspace Prompt Files

Stored in .github/prompts folder

✓ Only available within workspace

User Prompt Files

Stored in VS Code profile folder

✓ Available across all workspaces

Prompt File Structure

Extension: .prompt.md

YAML Header (optional):

- description, name, argument-hint
- agent (ask, edit, agent, custom)
- model (GPT-4o, Claude Sonnet 4, etc.)
- tools (list of available tools)

Body:

Contains prompt text sent to LLM with instructions and guidelines

Can include:

- Markdown links to workspace files
- Variables: ${workspaceFolder}, ${selection}
- Tool references: #tool:githubRepo
- Input variables: ${input:variableName}

Example: React Form

Complete prompt file structure

```
agent: 'agent'
model: GPT-4o
tools: ['githubRepo', 'search/codebase']
description: 'Generate a new React form component'
---
Your goal is to generate a new React form component based on the templates in #tool:githubRepo contoso/react-
templates.
Ask for the form name and fields if not provided.
Requirements for the form:
* Use form design system components: [design-system/Form.md](../docs/design-system/Form.md)
* Use `react-hook-form` for form state management:
* Always define TypeScript types for your form data
* Prefer *uncontrolled* components using register
* Use `defaultValues` to prevent unnecessary rerenders
* Use `yup` for validation:
* Create reusable validation schemas in separate files
* Use TypeScript types to ensure type safety
* Customize UX-friendly validation rules
```

Example: Example: perform a security review of a REST API

Complete prompt file structure

```
---
agent: 'ask'
model: Claude Sonnet 4
description: 'Perform a REST API security review'
---
Perform a REST API security review and provide a TODO list of security issues to address.
* Ensure all endpoints are protected by authentication and authorization
* Validate all user inputs and sanitize data
* Implement rate limiting and throttling
* Implement logging and monitoring for security events
Return the TODO list in a Markdown format, grouped by priority and issue type.
```

Create a Prompt File

Four simple steps

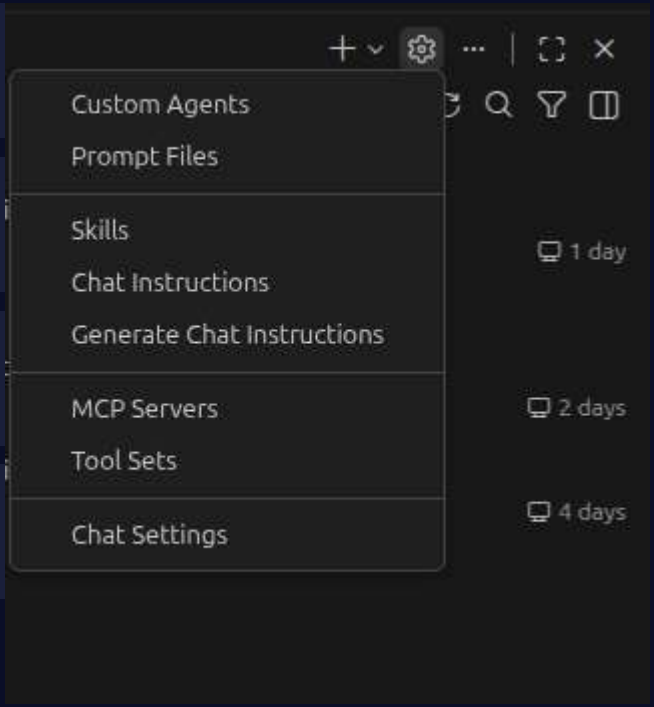
03

1. **Open Menu**
   Configure Chat > Prompt Files > New

2. **Choose Location**
   Workspace or User profile

3. **Enter File Name**
   Name appears when typing / in chat

4. **Author Prompt**
   Fill YAML frontmatter and body



Use Prompt Files

Three ways to run prompts

Type / in Chat

Type /prompt-name with optional parameters

/create-react-form formName=MyForm

Command Palette

Chat: Run Prompt > Select from list

Quick access to all prompts

Editor Play Button

Open prompt file > Click play in title bar

Test and iterate quickly

Tool List Priority

05

Order of tool selection

When both custom agent and prompt file specify tools

Priority Order:

1. Tools specified in prompt file (highest)
2. Tools from referenced custom agent
3. Default tools for selected agent (lowest)

Sync Across Devices

06

Keep prompts in sync with Settings Sync

User prompt files can sync across multiple devices

Setup Steps:

1. Enable Settings Sync in VS Code
2. Run Settings Sync: Configure command
3. Select 'Prompts and Instructions' to sync

Workspace prompts (.github/prompts) are not synced

Best Practices

Tips for effective prompts

Clear Instructions
Describe goals and expected output format

Provide Examples
Include input/output samples

Reference Custom Instructions
Use Markdown links vs duplicating

Use Variables
${selection} and input variables

Test with Play Button
Iterate and refine based on results

CUSTOM AGENTS IN VS CODE

AI personas tailored for specialized development roles

What Are Custom Agents?

AI personas with specific behaviors, tools, and instructions

An Agent Consists Of:

- Custom instructions defining behavior
- Specific set of available tools
- Tailored for specialized development roles
- Stored as .agent.md Markdown files

Example Personas:

Planner:	Generate implementation plans
Security Reviewer:	Audit code for vulnerabilities
Architect:	Design system architecture
Code Reviewer:	Review PRs and suggest improvements

Why Use Custom Agents?

01

Right tools for the right job

Task-Appropriate Tools

Planners use read-only, implementers use full editing

Specialized Instructions

Define exactly how AI should operate for each role

Consistent Responses

Same behavior every time you switch to that agent

Reusable Across Projects

Use in background and cloud agents too

Handoffs

02

Sequential workflows with suggested next steps

→ Transition between agents with context and pre-filled prompts

Example Workflows:

Planning → Implementation → Review

Write Failing Tests → Write Passing Tests

Architecture Design → Code Implementation

Security Audit → Fix Issues

How it works:

Handoff buttons appear after response → Click to switch agents with pre-filled prompt

Agent File Structure

03

Extension: .agent.md

YAML Header (optional):

- description, name, argument-hint
- tools (list of available tools)
- model (GPT-4o, Claude Sonnet 4, etc.)
- handoffs (label, agent, prompt, send)
- infer, target, mcp-servers

Body:

Contains instructions and guidelines for AI behavior in Markdown format

- Reference files with Markdown links
- Use tool references: #tool:githubRepo
- Prepended to user chat prompts
- Can reuse instruction files

Example: Planner Agent

Complete agent file structure

```
---
description: Generate implementation plan
name: Planner
tools: ['fetch', 'githubRepo', 'search']
model: Claude Sonnet 4
handoffs:
  - label: Implement Plan
    agent: agent
    prompt: Implement the plan outlined above.
---

# Planning instructions
Generate an implementation plan.
Don't make code edits, just create a plan with:
* Overview and requirements
* Implementation steps
* Testing strategy
```

Create a Custom Agent

04

Four simple steps

1. **Open Menu**
   Configure Custom Agents > Create new
2. **Choose Location**
   Workspace (.github/agents) or User profile
3. **Enter File Name**
   Name appears in agents dropdown
4. **Define Agent**
   Fill YAML header and body instructions

Tool List Priority

05

How tools are determined

Both custom agents and prompt files can specify tools

Priority Order:

1. Tools in prompt file (highest priority)
2. Tools from custom agent in prompt file
3. Default tools for selected agent (lowest)

Share Across Teams

06

Organization-level custom agents (Experimental)

User Level

Personal agents across all workspaces

User profile folder

Workspace Level

Team agents in .github/agents folder

Shared via repository

Organization Level

Org-wide agents for all members

GitHub organization (experimental)

Customize Dropdown

07

Show or hide agents in dropdown list

Control which custom agents appear in agents dropdown

Steps:

1. Select 'Configure Custom Agents' from dropdown
2. Hover over agent in list
3. Click eye icon to show/hide in dropdown

Use Case:

Hide extension-contributed agents without uninstalling the extension

Best Practices

Create effective custom agents

Clear Role Definition
Define specific purpose and behavior

Appropriate Tools
Read-only for planners, full edit for implementers

Use Handoffs
Create guided workflows between agents

Test and Iterate
Refine instructions based on results

Share with Team
Use workspace agents for consistency

HOW TO CREATE HANDOFFS

Step-by-step tutorial with examples

Before You Start

What you need

Requirements:

1. VS Code installed with GitHub Copilot extension
2. At least two custom agents created (.agent.md files)
3. Basic understanding of YAML format
4. Agent files in .github/agents folder (workspace) or user profile

Example Scenario We'll Build:

Planning Agent → Implementation Agent

User clicks button after planning to start implementation

Context flows: plan details → implementation prompt

Step 1: Locate Agent File

01

Find the agent file you want to add handoffs to

Method 1: Via Chat Interface

- Open Chat View in VS Code
- Click agents dropdown (top of chat)
- Select 'Configure Custom Agents'
- Click on your planning agent from the list

Method 2: Direct File Access

- Navigate to .github/agents/planning.agent.md
- Or your user profile agents folder

File Location Example:

```
.github/agents/planning.agent.md
```

Step 2: View Current File

02

See what the file looks like before adding handoffs

Original Planning Agent (BEFORE):

```
---
description: Generate an implementation plan
name: Planning Agent
tools: ['search', 'fetch', 'codebase']
model: Claude Sonnet 4
---

# Planning instructions
You are in planning mode. Generate a detailed
implementation plan including:
* Overview and requirements
* Step-by-step implementation guide
* Testing strategy

Do NOT make any code changes.
```

This agent works fine but has NO handoffs yet

Step 3: Add Handoffs Section

03

Add the handoffs field to YAML frontmatter

What to do:

1. Place cursor after the last field in YAML frontmatter (before ---)
2. Add a new line
3. Type: handoffs:
4. Press Enter to start defining handoff items

File with handoffs: added (NEW):

```
---
description: Generate an implementation plan
name: Planning Agent
tools: ['search', 'fetch', 'codebase']
model: Claude Sonnet 4
handoffs:
  - label: Start Implementation
    agent: implementation
    prompt: Now implement the plan above.
    send: false
---
```

Step 4: Define Handoff Properties

04

Fill in the four required fields

label: Text shown on the button

Example: 'Start Implementation'

Tip: Use action verbs, keep it short

agent: Target agent identifier

Example: 'implementation'

Tip: Must match agent name exactly

prompt: Message sent to next agent

Example: 'Now implement the plan outlined above.'

Tip: Reference previous work

send: Auto-submit the prompt?

Example: false (default: user reviews first)

Tip: Use false for safety

Step 5: Complete Example

05

Full agent file with handoff

```
---
description: Generate an implementation plan
name: Planning Agent
tools: ['search', 'fetch', 'codebase']
model: Claude Sonnet 4
handoffs:
  - label: Start Implementation
    agent: implementation
    prompt: |
      Now implement the plan outlined above.
      Follow the architecture decisions and
      use the recommended libraries.
    send: false
---

# Planning instructions
You are in planning mode. Generate a detailed
implementation plan. Do NOT make code changes.
```

Step 6: Save and Test

06

Save file and test the handoff

Save the File

Press Ctrl+S (Windows/Linux) or Cmd+S (Mac)

Testing Steps:

1. Open Chat View in VS Code
2. Switch to your Planning Agent
3. Ask it to create a plan (e.g., 'Plan a login feature')
4. Wait for response to complete
5. Look for the 'Start Implementation' button below response
6. Click the button - you should switch to Implementation agent
7. Verify the prompt is pre-filled with your message

Adding Multiple Handoffs

07

Offer multiple next steps

You can add multiple handoff options - just add more items to the list

```yaml
handoffs:
  - label: Start Implementation
    agent: implementation
    prompt: Implement the plan above.
    send: false
  - label: Review Architecture
    agent: architect
    prompt: Review this plan for issues.
    send: false
  - label: Create Tests First
    agent: test-writer
    prompt: Write tests based on this plan.
    send: false
```

Result: Three buttons appear after planning response

Common Mistakes to Avoid

Watch out for these errors

✗ Wrong indentation in YAML

✓ Fix: Use exactly 2 spaces for each level, no tabs

```
  - label: (correct) vs - label: (wrong)
```

✗ Agent name doesn't match

✓ Fix: Agent identifier must match exactly (case-sensitive)

```
'implementation' not 'Implementation Agent'
```

✗ Missing quotes in strings

✓ Fix: Wrap text values in single or double quotes

```
label: 'Start' not label: Start
```

✗ Handoffs outside YAML section

✓ Fix: Must be inside --- markers at top of file

```
Place before closing --- line
```

Troubleshooting

What to do when things don't work

Problem: Handoff button doesn't appear

Solutions:

- Check YAML syntax is correct (use YAML validator)
- Ensure file is saved (Ctrl+S / Cmd+S)
- Restart VS Code if needed
- Verify agent completed its response fully

Problem: Button appears but nothing happens

Solutions:

- Check target agent name matches exactly
- Ensure target agent exists and is accessible
- Look for errors in VS Code Developer Tools
- Try with send: false first for debugging

Pro Tip: Test with send: false first, then switch to true when working

Quick Reference

Handoff template to copy

```yaml
handoffs:
  - label: 'Your Button Text Here'
    agent: 'target-agent-name'
    prompt: 'Your message to next agent'
    send: false
```

Remember:

- ✓ label: Text shown on button (action-oriented)
- ✓ agent: Must match target agent name exactly
- ✓ prompt: Message with context for next agent
- ✓ send: false (review first) or true (auto-submit)

Practice Exercise

Try creating this handoff

Exercise: Implementation → Review Workflow

After implementation completes, add a handoff button that:

1. Shows button labeled 'Review My Code'
2. Switches to a 'code-reviewer' agent
3. Sends message: 'Please review the code I just implemented'
4. Requires user to review prompt before sending

Solution on next slide (try first!)

Hints:

- Remember to use quotes around text values
- Agent name is 'code-reviewer' (lowercase, with hyphen)
- send: false means user reviews before sending
- Place in implementation.agent.md file

Exercise Solution

How did you do?

```
---
description: Implement code based on plan
name: Implementation Agent
tools: ['edit', 'codebase', 'terminal']
model: Claude Sonnet 4
handoffs:
  - label: 'Review My Code'
    agent: 'code-reviewer'
    prompt: 'Please review the code I just implemented'
    send: false
---

# Implementation instructions
Implement the planned features with full editing tools.
```

This creates a handoff from implementation to code review!

AGENT SKILLS

Specialized capabilities with instructions and resources

What Are Agent Skills?

Folders with instructions, scripts, and resources for specialized tasks

A Skill Contains:

- SKILL.md file with instructions
- Scripts and automation tools
- Example files and templates
- Documentation and resources
- All stored in a single directory

Key Benefits:

- Specialize Copilot for domain-specific tasks
- Reduce repetition across conversations
- Compose multiple skills for complex workflows
- Load only relevant content when needed

Skills vs Custom Instructions

Different tools for different purposes

01

Purpose

Skills: Specialized capabilities and workflows

Instructions: Coding standards and guidelines

Portability

Skills: VS Code, CLI, Copilot coding agent

Instructions: VS Code and GitHub.com only

Content

Skills: Instructions, scripts, examples, resources

Instructions: Instructions only

Scope

Skills: Task-specific, loaded on-demand

Instructions: Always applied (or via glob patterns)

Standard

Skills: Open standard (agentskills.io)

Instructions: VS Code-specific

When to Use Skills

02

Choose the right tool for your needs

Use Agent Skills When:

- ✓ Creating reusable capabilities across AI tools
- ✓ Including scripts, examples, or resources
- ✓ Sharing capabilities with the AI community
- ✓ Defining specialized workflows (testing, debugging, deployment)

Use Custom Instructions When:

- ✓ Defining project-specific coding standards
- ✓ Setting language or framework conventions
- ✓ Specifying code review or commit guidelines
- ✓ Applying rules based on file types with glob patterns

Skill Directory Structure

03

Where skills are stored

Project Skills

`.github/skills/`

Stored in repository, shared with team

Personal Skills

`~/.copilot/skills/`

Stored in user profile, available across projects

Example Skill Directory:

```
.github/skills/webapp-testing/
  ├── SKILL.md              # Main instructions
  ├── test-template.js      # Template file
  └── examples/             # Example scenarios
      ├── login-test.js
      └── checkout-test.js
```

SKILL.md File Format

Two parts: YAML header and markdown body

YAML Header (Required):

`name`

Unique identifier (lowercase, hyphens, max 64 chars)

`description`

What it does AND when to use it (max 1024 chars)

Markdown Body:

Contains instructions that describe:

- What the skill helps accomplish
- When to use the skill
- Step-by-step procedures to follow
- Examples of expected input and output
- References to included scripts or resources

Example: Testing Skill

Complete SKILL.md file

```
name: webapp-testing
description: Guide for testing web applications using Playwright. Use this when asked to create or
run browser-based tests.
---
# Web Application Testing with Playwright
This skill helps you create and run browser-based tests for web applications using Playwright.
## When to use this skill
Use this skill when you need to:
- Create new Playwright tests for web applications
- Debug failing browser tests
- Set up test infrastructure for a new project
## Creating tests
1. Review the [test template](./test-template.js) for the standard test structure
2. Identify the user flow to test
3. Create a new test file in the `tests/` directory
4. Use Playwright's locators to find elements (prefer role-based selectors)
5. Add assertions to verify expected behavior
## Running tests
To run tests locally:
```bash
npx playwright test
```
To debug tests:
```bash
npx playwright test --debug
```
Best practices
- Use data-testid attributes for dynamic content
- Keep tests independent and atomic
- Use Page Object Model for complex pages
- Take screenshots on failure
```

How Copilot Uses Skills

Progressive disclosure: three-level loading

1

Skill Discovery

Reads name and description from all skills

When: Always available, lightweight metadata

2

Instructions Loading

Loads SKILL.md body into context

When: When request matches skill description

3

Resource Access

Accesses scripts, examples, documentation

When: Only when Copilot references them

Skills activate automatically based on your prompt - no manual selection needed

Create a Skill

Four simple steps

06

1. **Create Directory**
   Make .github/skills directory in workspace

2. **Create Subdirectory**
   One directory per skill (e.g., webapp-testing)

3. **Create SKILL.md**
   Add YAML header (name, description) and body

4. **Add Resources (Optional)**
   Include scripts, templates, examples

Use Shared Skills

Leverage community-created skills

Community Repositories:

github/awesome-copilot- https://github.com/github/awesome-copilot

Skills, agents, instructions, and prompts

anthropics/skills - https://www.aitmpl.com/hooks

Reference skills collection

How to Use Shared Skills:

1. Browse available skills in repository
2. Copy skill directory to your .github/skills/ folder
3. Review and customize SKILL.md for your needs
4. Optionally modify or add resources

Always review shared skills before use for security and quality

Open Standard

Portable across multiple AI agents

Agent Skills is an open standard at agentskills.io

Works With Multiple Agents:

GitHub Copilot in VS Code

Available in chat and agent mode

GitHub Copilot CLI

Accessible when working in terminal

GitHub Copilot coding agent

Used during automated coding tasks

Key Benefit: Write once, use everywhere

Best Practices

Create effective skills

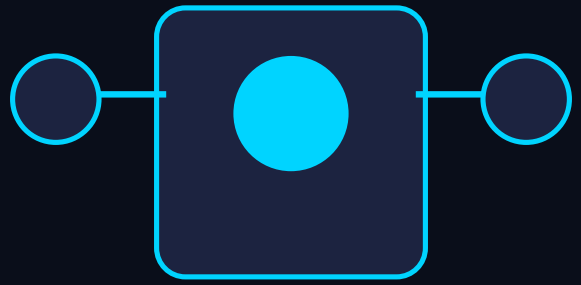
Clear Descriptions
Explain what AND when - helps Copilot decide when to load

Step-by-Step Instructions
Provide clear procedures with examples

Include Resources
Add templates, scripts, and examples

Test Your Skills
Verify they work with different prompts

Share with Community
Contribute to awesome-copilot repository



MODEL CONTEXT PROTOCOL

The universal standard for connecting AI to everything

What is MCP?

MCP

An open-source standard for connecting AI applications to external systems

Key Capabilities:

- ✓ Connect to data sources — local files, databases, cloud storage
- ✓ Integrate with tools — search engines, calculators, APIs
- ✓ Access specialized workflows and AI-powered prompts
- ✓ Enable agents to access information and perform actions
- ✓ Works with Claude, ChatGPT, and other AI applications

Think of MCP like a USB-C port for AI — one universal standard for everything

Why MCP Matters

02

Benefits across the ecosystem

DEV

Developers

Reduces development time and complexity when building AI applications or integrating external tools

AI

AI Agents

Provides access to an ecosystem of data sources, tools and apps that enhance capabilities and user experience

USR

End-Users

Results in more capable AI that can access your data and take actions on your behalf automatically

Without MCP:

Every AI integration required custom, one-off connections — complex and hard to maintain

How MCP Works

Architecture overview

Servers provide:

- Tools — actions AI can invoke (e.g. search, write file)
- Resources — data AI can read (e.g. docs, DBs)
- Prompts — reusable workflow templates

Clients can use:

- Discover available tools and resources
- Execute tool calls on behalf of the AI
- Manage context and conversation flow

Real-World Examples

What MCP can enable today

CALENDAR + NOTES

Personal AI Assistant

Agents access Google Calendar and Notion, becoming a fully personalized AI assistant that knows your schedule and notes

FIGMA → CODE

Design to Code

Claude Code generates an entire web app directly from a Figma design file — bridging design and development workflows

DATABASES

Enterprise Data Access

Enterprise chatbots connect to multiple databases across an organization, enabling data analysis via natural language

BLENDER + PRINTER

3D Design & Print

AI models create 3D designs in Blender and send them directly to a 3D printer — physical world automation

Start Building with MCP

Three paths to get started

1

Build an MCP Server

Create an MCP server to expose your data sources and tools. Use the official SDKs for Python, TypeScript, or other supported languages.

2

Build an MCP Client

Develop an application that connects to MCP servers. Your client can discover tools and resources, then invoke them on behalf of users.

3

Connect & Explore

Use the MCP Inspector to debug and test your integrations. Browse the MCP Registry to find existing servers and connect to them.

docs: modelcontextprotocol.io/docs/getting-started/intro